

XUANZHE HAN

+1 540-840-9286 ◇ hanxuanzhe2002@gmail.com

linkedin.com/in/xuanzhe-han-542225206 ◇ xuanzhehan.netlify.app

EDUCATION

Master of Science in Computer Science

2024 - present

University of California, Los Angeles

Bachelor of Science in Computer Science

2020 - 2024

University of California, Los Angeles

SKILLS

Languages: C/C++, Python, Java, JavaScript, HTML/CSS, TypeScript, SQL, Scheme/LISP, OCaml

Frameworks & Libraries: React, ROS, PyTorch, Flask, FastAPI, CUDA, Scikit-learn

Tools & Databases: MongoDB, Web Storage API, Docker

EXPERIENCE

Systems Research Intern

Sep 2024 – Mar 2025

Nissan Group North America, Inc.

- Aided in the development of heuristics for constraint-based routing.
- Built a customizable ROS + C++ data pipeline to process sensing data into ML-ready inputs, enabling context-aware, time-accurate routing (pedestrian density, AV speed, road obstructions, etc.).
- Helped in the development of data indexing tools for AV research.

Web Development Intern

Oct 2023 – Jan 2024

Tap Series, LLC

- Designed and implemented a React + Flask visualization tool to parse sales/customer interaction data and display trends by time, region, and demographics.
- Supported team projects by updating legacy web apps and adding admin-focused tools, improving decision-making workflows.
- Integrated Twilio SendGrid REST API for automated communication and reporting.

Research Intern, Facial Action Unit Detection

Aug – Sep 2023

Huazhong University of Science and Technology

- Adapted JAANet and other ML models to new facial expression datasets, providing standardized baselines for ongoing research.
- Implemented and benchmarked PyTorch and scikit-learn pipelines to evaluate action unit detection performance.
- Contributed to literature reviews and experimental documentation for publication prep.

RELATED COURSEWORK

Systems & Networks: Operating Systems, Compilers, Computer Architecture, Parallel & Distributed Computing

AI & Data: Machine Learning, NLP, Computer Vision, Data Management Systems

Software Development: Design of Programming Languages, Algorithms & Complexity, Web Application Development, Networking Fundamentals, Microservices

EXTRACURRICULAR ACTIVITIES

Webmaster & Team Lead, 3D4E (3D Printing Club)

2022 – 2024

- Maintained and updated club website (3d4eatucla.github.io) to improve usability and member engagement.
- Led biomedical and mechanical keyboard design teams, delivering functional prototypes including oximeters and custom macro pads.

PROJECTS

P-Forest

Python, Scikit-learn

Implemented a network intrusion detection system based on "*p-forest: In-Network Inference with Random Forests*". Developed vanilla and compressed P-Forest implementations, generated TCAM/CRAM rules and RMT simulator inputs, and built visualization tools to benchmark performance on the CIC-IDS-2017 dataset.

github.com/HiccupHan/CS216-PForest-Project

Dynamic Kafka

Python, Docker, Kafka

Prototyped dynamic batching in Apache Kafka by containerizing brokers, producers, and consumers with Docker Compose. Built Python producers with adjustable batching logic, automated topic management scripts, and explored auto-tuning for latency-throughput tradeoffs.

github.com/Hersh245/dynamic-kafka

CUDA RNN

C++, CUDA

Implemented a recurrent neural network (RNN) from scratch with CUDA for GPU acceleration. Optimized matrix ops and memory usage for parallelized training, achieving significant speedups over CPU-only baselines.

github.com/HiccupHan/CS-259-CUDA-RNN

Cache System

Python, FastAPI

Designed a microservice framework to evaluate caching strategies. Implemented baseline, prefetch, and tiered caching, analyzing tradeoffs in latency, memory, and hit rate.

github.com/lcptgeneral/cs239-caching

Recommendation System

Python

Explored probabilistic models for user preference prediction. Implemented and compared ALS-PMF, SGD-PMF, and LDA in a modular experimentation framework.

github.com/HiccupHan/Recommendation-System

SlangLingo

React, OpenAI API

Developed a translation web app that handles tricky slang with LLMs.

github.com/HiccupHan/SlangLingo

ResuMatch

Chrome Extension, JavaScript

Created a Chrome extension that matches resumes to LinkedIn postings to streamline applications.

github.com/HiccupHan/ResuMatch